

Data Collection, Instruments and Measurement

120 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Turning abstract constructs into things you can actually record, and the validity and reliability evidence needed before anyone believes the numbers your instrument produces.

Learning objectives

- Operationalise a construct into measurable variables and indicators
- Identify levels of measurement and their analytical consequences
- Design questionnaire items that avoid the common wording faults
- Distinguish the main types of validity and reliability and how each is evidenced
- Plan and report a pilot study
- Choose between paper and electronic data capture and set up basic data quality controls

From construct to variable

- A construct is an abstraction — stigma, food insecurity, quality of care — that cannot be observed directly
- Operationalisation defines the observable indicators that will stand for it
- A conceptual definition says what you mean; an operational definition says exactly how it will be measured
- Prefer an existing validated instrument over inventing your own — validation takes a study of its own
- If you adapt an instrument, report what you changed and re-examine its reliability in your population
- Instruments validated elsewhere may behave differently after translation or in a different culture, and should be re-tested

IN PRACTICE

Translation requires forward translation, back translation by an independent translator, reconciliation by a panel, and cognitive interviewing with a few target participants. Translating a questionnaire yourself the night before is the single most common source of unusable data.

Levels of measurement

- Nominal — unordered categories (blood group, district); only counts and proportions are meaningful

- Ordinal — ordered categories with unequal or unknown intervals (Likert responses, disease stage)
 - Interval — equal intervals but no true zero (temperature in Celsius)
 - Ratio — equal intervals with a true zero (weight, income, count of visits)
 - The level determines which summary statistics and which tests are legitimate
 - Record data at the finest level available: exact age can always be collapsed into bands later, but bands can never be recovered into exact ages
-

Writing good items

- One idea per item — "Was the staff friendly and efficient?" is double-barrelled and unanswerable if one was and one was not
 - Avoid leading wording, loaded terms and negatives, especially double negatives
 - Match vocabulary to the least literate respondent you expect
 - Make response options exhaustive and mutually exclusive; check for gaps and overlaps in age bands
 - Offer "don't know" and "prefer not to answer" where forcing a choice would fabricate data
 - Decide deliberately whether a Likert scale has a midpoint — removing it forces a direction, which may or may not be what you want
 - Order matters: put sensitive items late, once rapport exists, and beware of earlier items priming later ones
-

Validity and reliability

- Face validity — does it look appropriate? The weakest form, but worth checking
- Content validity — does it cover the whole construct? Established by expert panel, often with a content validity index
- Construct validity — does it behave as theory predicts? Evidenced by convergent and discriminant correlations and factor analysis
- Criterion validity — does it agree with a gold standard, concurrently or predictively?
- Internal consistency — do items measure the same thing? Cronbach's alpha, where roughly 0.70-0.95 is acceptable and above 0.95 suggests redundant items
- Test-retest reliability — does it give the same answer on a stable characteristic over a short interval?
- Inter-rater reliability — do different observers agree? Reported as Cohen's kappa or an intraclass correlation
- An instrument can be reliable but not valid — consistently measuring the wrong thing

WHICH EVIDENCE TO REPORT

Property	Statistic	Rough benchmark
Internal consistency	Cronbach's alpha	0.70 - 0.95
Test-retest	ICC or Pearson r	≥ 0.70
Inter-rater agreement	Cohen's kappa	≥ 0.60 substantial
Content validity	CVI	≥ 0.78 per item
Construct validity	Factor loadings	≥ 0.40

Piloting and data capture

- Pilot on 10% of the intended sample, or at least 20-30 respondents, drawn from the same population but excluded from the main study
- The pilot tests comprehension, timing, flow, skip logic, coding and the analysis plan — run your intended analysis on pilot data
- Electronic capture (KoBoToolbox, REDCap, ODK, CommCare) enforces ranges and skip logic at the point of entry and removes a transcription step
- Paper remains appropriate where power, connectivity or literacy make devices impractical; plan double entry if so
- Build validation rules: allowed ranges, required fields, logical consistency checks between related items
- Train and standardise data collectors, then check inter-rater agreement rather than assuming it
- Anonymise at collection where possible, keeping any identifier link in a separate secured file

Worked examples

A questionnaire item pulled apart

A draft patient satisfaction survey includes: "How satisfied were you with the cost and quality of care? (Very satisfied / Satisfied / Dissatisfied)"

1. Double-barrelled: a patient happy with quality but not cost cannot answer truthfully.
2. The response options are unbalanced — two positive, one negative — which pushes answers upward.
3. There is no neutral option and no "not applicable", so someone who paid nothing must still choose.
4. Split into two items, each with a balanced 5-point scale from Very dissatisfied to Very satisfied.
5. Add "Not applicable" to the cost item for patients covered by the NHIS.
6. Pilot on 20 patients and ask them to say aloud what they thought each item meant.

THE POINT

Every faulty item produces data you cannot interpret, and no analysis later can repair it. An hour spent on wording saves the whole variable.

What a translated instrument really requires

A validated English depression scale is to be used with Twi-speaking participants. The student translates it herself the week before fieldwork.

1. Forward translation by two independent bilingual translators, working separately.
2. Reconciliation of the two versions by a panel, resolving each difference deliberately.
3. Back translation into English by a third translator who has never seen the original.
4. Compare back translation with the original; where meaning has drifted, revise and repeat.
5. Cognitive interviewing with 5-10 target participants: ask what each item means to them in their own words.
6. Re-check reliability in the new population — the original Cronbach's alpha does not transfer.

THE POINT

A self-translated instrument has unknown validity, which means the study built on it does too. This is one of the most common fatal flaws in multilingual research.

Key terms

Specifying the observable indicators and procedures by which an abstract construct will be measured.

Construct validity

Evidence that an instrument measures the theoretical concept it claims to measure.

Cronbach's alpha

A coefficient of internal consistency among the items of a scale.

Cohen's kappa

A measure of agreement between raters that corrects for agreement expected by chance.

Skip logic

Rules routing a respondent past items that do not apply to them.

Double entry

Entering paper data twice independently and reconciling differences to catch transcription error.

Activity

Take one construct from your study and write three candidate items for it. Exchange with a partner, who must identify any double-barrelled, leading or ambiguous wording and check the response options are exhaustive and mutually exclusive. Then state which reliability evidence you would report and why.

Further reading

- **DeVellis — Scale Development: Theory and Applications** — The standard reference for constructing and validating scales.
- **Boateng et al. (2018) — "Best Practices for Developing and Validating Scales"** — Open access, health-focused, and unusually practical.
- **Beaton et al. (2000) — Guidelines for cross-cultural adaptation** — The forward/back translation process, done properly.

Self-check questions

1. "How satisfied were you with the cost and quality of care?" is faulty because it is:
 - a. Leading
 - b. Double-barrelled
 - c. Too open
 - d. Correctly written

2. A scale consistently gives the same score on repeat administration but correlates poorly with every established measure of the construct. It is:
 - a. Valid but unreliable
 - b. Reliable but not valid
 - c. Both reliable and valid
 - d. Neither
 3. Recording age in five-year bands rather than exact years is generally inadvisable because:
 - a. Bands are harder for respondents to answer
 - b. Exact values can always be collapsed into bands later, but bands cannot be recovered
 - c. Bands are not a valid level of measurement
 - d. It always reduces the response rate
 4. A Cronbach's alpha of 0.98 on a 20-item scale most likely indicates:
 - a. Excellent construct validity
 - b. Redundant items that are near-duplicates of one another
 - c. Poor internal consistency
 - d. A sample that is too small
 5. The primary purpose of a pilot study is to:
 - a. Increase the total sample size
 - b. Test comprehension, timing, flow, coding and the analysis plan before full data collection
 - c. Publish preliminary findings
 - d. Satisfy the ethics committee
 6. When adapting a validated English instrument for use in Twi, the minimum acceptable process includes:
 - a. Direct translation by the researcher
 - b. Forward translation, independent back translation, expert reconciliation and cognitive interviewing
 - c. Using an online translation tool and checking it reads well
 - d. No adaptation is needed if respondents understand some English
-

Answers

1. **b.** It asks about two distinct things at once. A respondent satisfied with quality but not cost cannot answer, and their response cannot be interpreted. Split it into two items.

2. **b.** Consistency is reliability. Failing to relate to the construct it claims to measure is a validity failure. An instrument can measure the wrong thing very consistently.
3. **b.** Collecting at the finest available level preserves analytical options. Categorising at collection discards information permanently and forecloses analyses you may later need.
4. **b.** Very high alpha usually means items are asking essentially the same question in different words. It inflates apparent consistency while narrowing the construct actually covered.
5. **b.** A pilot is a rehearsal of the whole process, including running the intended analysis on pilot data. Pilot participants are excluded from the main study and do not add to its sample.
6. **b.** Cross-cultural adaptation is a defined procedure. Skipping it means the translated instrument's validity is unknown, and any psychometric evidence from the original version no longer applies.

